

Existence, Localization, and Stability of a Limit Cycle in a Class of Benchmark Biomolecular Oscillators

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Oscillatory behaviour is important in multiple biological contexts. However, the inherent nonlinearity and high dimensionality of mathematical models in biology makes proving the existence and the localization of limit cycle oscillations challenging. Here, we provided an elementary proof for the existence and a method for rigorously localizing the oscillatory solutions in a class of benchmark biomolecular oscillators. To construct the proof, we used a geometric approach based on Brouwer’s Fixed Point theorem. We constructed a toroidal-like manifold within a positively invariant set by removing the hypervolume containing the fixed point and the trajectories converging to it along its stable manifold. We showed that the vector field describing the system dynamics maps a cross section of the toroidal-like manifold onto itself. The existence of a limit cycle solution in this manifold was guaranteed by Brouwer’s Fixed Point theorem. For different sets of initial conditions in these cross-sections, we used an interval-based Reachability Analysis to localize the oscillatory solution. The stability of the limit cycle was established through an interval-based Poincaré Map analysis. The Jacobian of the discrete Poincaré Map was calculated over the localized cross-section using the augmented state-variational system. The sub-elements of the previously localized region were identified where the eigenvalues of the Jacobian of the Poincaré Map lay within the unit circle, guaranteeing contraction of the return map and hence stability. These results add a simple and elegant approach to demonstrating the existence of limit cycles in biomolecular systems, a method for rigorous localization of the region of existence, as well as an interval-based Poincaré Map analysis to show the stability of the limit cycle.

We presented an elementary proof for the existence of oscillatory solutions and a rigorous method to localize them in a class of benchmark biomolecular oscillators. First, we used a geometric approach to construct a toroidal-like manifold that is positively invariant by removing the region containing the fixed point and its stable manifold. Second, we then showed that the vector field due to the system dynamics maps a cross-section onto itself, guaranteeing the existence of a limit cycle based on Brouwer’s Fixed Point theorem. Third, we used a rigorous interval-based Reachability Analysis to localize the oscillatory behavior for different initial conditions, complementing the topological arguments of Brouwer’s Fixed Point theorem. Fourth, we established the stability of the localized limit cycle using an interval-based Poincaré Map analysis. The Jacobian of the discrete Poincaré Map was rigorously enclosed using the variational dynamics, and contraction condition was verified over the identified sub-elements. This contraction guarantees that the corresponding Poincaré Map is locally contractive, thereby establishing the stability of the limit cycle.

cycle is not needed. This theorem is not applicable in dimensions greater than two because of the possibility of more complex attractors and chaotic behaviour. Complementing this, recent works on the nonexistence criteria in planar systems act as a filter to rule out closed orbits in specific regions narrowing the search for the invariant sets^{2,3}. This significantly improves efficiency by identifying one-dimensional manifolds that must intersect any existing cycle, allowing for targeted seeding of initial conditions rather than a computationally expensive search^{2,3}.

Limit cycles in systems of dimension greater than two, however, are typically found in biological systems, where they underlie crucial control processes such as circadian rhythms and the cell cycle⁴. For example, the famous Goodwin oscillator has a negative feedback loop wrapping a chain of intermediate biomolecules so that the overall length of the chain is greater than or equal to three^{5–7}. Another canonical example is a ring oscillator with an odd number (≥ 3) of genes that sequentially repress each other⁸. The experimental demonstration of the repressilator, a ring oscillator with three genes, heralded a new era for synthetic biology⁹. Therefore, methods to establish the existence and rigorous enclosure of limit cycles in higher-dimensional systems are needed.

While various approaches have been used to prove the existence of limit cycles in biological systems of dimensions greater than or equal to three, the region of existence is relatively large and conservative. Like the Poincaré-Bendixson theorem, these approaches seek to establish the existence of a limit cycle without an explicit analytical solution. One approach is to analyze biological oscillators in a Boolean framework using kinetic logic models^{8,10}. A geometric ap-

I. INTRODUCTION

The Poincaré-Bendixson theorem provides a way to demonstrate the existence of limit cycle oscillations in two-dimensional systems¹. The basic idea is to construct a positively invariant set without steady states so that the trajectories converge to the limit cycle. An explicit solution of the limit

proach based on Brouwer's fixed point theorem was used to prove the existence of a limit cycle in the three-dimensional Goodwin oscillator¹¹, which we recently adapted for the repressilator¹². This approach was subsequently generalized to an n -dimensional Goodwin oscillator¹³. This approach has been referred to as 'elaborate'¹⁴ and 'laborious'¹⁵, but provides concrete insight into the limit cycle behaviour. An n -dimensional version of the Poincare-Bendixson Theorem was developed for monotone cyclic feedback systems¹⁶. Moreover, it has been demonstrated that the linearisation at a stationary solution results in a global Hopf bifurcation when the coefficients along such a cycle are sufficiently dominant¹⁷. The above ideas were used to address the existence and stability in ring oscillators and their variants^{14,18}. An explicit characterization of biological parameter sets for the existence of oscillations for the ring oscillators was also reported¹⁹⁻²¹. Recent work has applied the notion of 2-cooperativity to prove the existence of limit cycles with wide applicability in biology²²⁻²⁴. Typical estimates of the region of existence of the limit cycle obtained using the above approaches are conservative in nature, and rigorous tighter enclosures are desirable.

In this work, we localized the region of existence of the limit cycle in a benchmark biomolecular oscillators using the Brouwer's Fixed Point theorem and interval-based Reachability Analysis. We provided an alternative, elementary proof for the existence of a limit cycle in cyclic gene regulatory network with odd number of genes. For the existence proof, we constructed a positively invariant set for oscillator dynamics by analyzing the direction of the vector field at the outer facets of a hypercube. We constructed a torus-like structure by removing a hypervolume associated with the unstable steady state and the trajectories approaching it along its stable manifold. We showed that a cross-section of this torus-like region maps onto itself, thus proving the existence of a limit cycle. For different sets of initial conditions on these cross-sections, we rigorously evolved the network dynamics using Interval Analysis to localize the limit cycle. Further in this localized region of the cross-sections, we used an interval-based Poincaré Map to certify the stability of the limit cycle.

II. EXISTENCE

In this section, an elementary proof for the existence of a limit cycle in m -gene cyclic regulatory network based on Brouwer's Fixed Point theorem is presented.

A. Methodology

Let $x_1, x_2, \dots, x_m$ represent the proteins of the m -gene cyclic regulatory network. The differential equations describing the dynamics of the network are given in (2). A symmetric model with only the protein dynamics, ignoring mRNA dynamics, is chosen for simplicity.

$$\dot{x} = f(x), \quad (1)$$

$$\implies \frac{dx_i}{dt} = F(x_{i-1}) - \gamma x_i, \text{ for } i = 1, 2, 3, \dots, m. \quad (2)$$

Here x_i is the concentration of the i^{th} protein with $0 \leq x_i \leq \frac{\alpha}{\gamma}$. α, γ, k , and n are positive constants, $F(p) = \frac{\alpha}{1 + (\frac{p}{k})^n}$ is the Hill function, n is the Hill coefficient, and m is an odd number. As the system is cyclic in nature, $x_0 = x_m$.

The steady state of the cyclic regulatory network in (2) is unique. The uniqueness can be demonstrated graphically⁴. Denote the steady state as $(x_{10}, x_{20}, \dots, x_{m0})$. So, $x_{10} = \frac{1}{\gamma} F\left(\underbrace{\frac{1}{\gamma} F \dots \left(\frac{1}{\gamma} F(x_{10})\right) \dots}_{m\text{-times}}\right)$. As the gene regulatory net-

work model is symmetric, the steady state is of the form $(x_0, x_0, \dots, x_0)$. The stability of this steady state can be assessed through the eigenvalues of the Jacobian of the linearization of the system in (2) at the steady state $(x_0, x_0, \dots, x_0)$. The eigenvalues of the Jacobian are: $\lambda_{1,\dots,m} = -\gamma + \rho \left[\cos\left(\frac{2p+1}{m}\pi\right) + j \sin\left(\frac{2p+1}{m}\pi\right) \right]$, where $p = 0, 1, \dots, m-1$ and $\rho = -F'(x_0)$. An eigenvalue is real and negative for $p = \frac{m-1}{2}$, $\lambda = -(\gamma + \rho)$. The remaining eigenvalues are complex in nature. For certain values of p , the cosine component in the real part of the eigenvalues become negative. This implies that the corresponding eigenvalues will certainly have negative real parts. We used the instability of the steady state to prove the following sufficient condition for the existence of a limit cycle.

Theorem 1. *There exists a limit cycle solution to the cyclic gene regulatory network in (2) if $\gamma < \max_{p \in \mathcal{P}_+} \left\{ \rho \cos\left(\frac{2p+1}{m}\pi\right) \right\}$, where $\mathcal{P}_+ = \left\{ p \in [0, m-1] : \cos\left(\frac{2p+1}{m}\pi\right) > 0 \right\}$.*

Proof. For certain values of p , the cosine component in the real part of the eigenvalues becomes negative. This implies that the corresponding eigenvalues will certainly have a negative real part. This contributes to the possible stability of the steady state. However, the instability of the steady state implies that nearby trajectories are repelled from the steady state and may converge to a closed invariant set, such as a limit cycle. To ensure that the steady state is unstable, at least one of the complex conjugate pair of eigenvalues should have a positive real part. For the complex eigenvalues to have a positive real part, $-\gamma + \rho \cos\left(\frac{2p+1}{m}\pi\right) > 0 \quad \forall p \in \mathcal{P}_+$, where, $\mathcal{P}_+ = \left\{ p \in [0, m-1] : \cos\left(\frac{2p+1}{m}\pi\right) > 0 \right\}$. So, to ensure at least one of the complex conjugate pair of eigenvalues has positive real part: $\gamma < \max_{p \in \mathcal{P}_+} \left\{ \rho \cos\left(\frac{2p+1}{m}\pi\right) \right\}$.

Upon establishing the instability of the steady state, a geometric approach is used to prove the existence of a limit cycle within the hypercube $0 \leq x_1, x_2, \dots, x_m \leq \frac{\alpha}{\gamma}$.

Construction of an Invariant Box: To demonstrate the existence of limit cycle in the m -gene cyclic regulatory network described by (2), we showed that the hypercube is a positively

invariant set. This is done by computing the inner product (Ξ) between the vector field characterized by the system dynamics within the hypercube ($\vec{G}$) and the normal to the outer facets of the hypercube ($\hat{n}$): $\Xi = \langle \vec{G}, \hat{n} \rangle$. We considered the unit normal vectors for the different facets of the hypercube directed inward. The value of Ξ determines the direction of the flow of the vector field through the facets: (i) $\Xi > 0$ indicates the vector field flows through the facet in the direction of $\hat{n}$, (ii) $\Xi = 0$ indicates the vector field does not flow through the facet, and (iii) $\Xi < 0$ indicates the vector field flows through the facet in the direction opposite of $\hat{n}$.

The vector field described by the system dynamics in (2) is: $\vec{G}(x_1, x_2, x_3, \dots, x_m) = [F(x_m) - \gamma x_1] \hat{x}_1 + [F(x_1) - \gamma x_2] \hat{x}_2 + [F(x_2) - \gamma x_3] \hat{x}_3 + \dots + [F(x_{m-1}) - \gamma x_m] \hat{x}_m$, where $\hat{x}_i$ (for $i = 1, 2, \dots, m$) are the unit vectors in the direction of x_i axis. We calculated the inner product (Ξ) of the vector field with unit normal vector to the different facets. The normal unit vector is always considered to be directed into the hypercube. The outer facets of the hypercube are $x_1 = 0$, $x_1 = \frac{\alpha}{\gamma}$, $x_2 = 0$, $x_2 = \frac{\alpha}{\gamma}$, $\dots$, $x_m = 0$, and $x_m = \frac{\alpha}{\gamma}$. We computed the inner product between the vector field characterized by the system dynamics within the hypercube and the normal to the outer facets of the hypercube.

Table I. Inner product between the vector field and the normal to the outer facets of the hypercube.

Facet	Unit normal Vector, $\hat{n}$	Inner Product, $\Xi = \langle \vec{G}, \hat{n} \rangle$	Value of Ξ
$x_i = 0$	$\hat{x}_i$	$\frac{\alpha}{1 + \left(\frac{x_{i-1}}{k}\right)^n}$	+ve
$x_i = \frac{\alpha}{\gamma}$	$-\hat{x}_i$	$\alpha - \frac{\alpha}{1 + \left(\frac{x_{i-1}}{k}\right)^n}$	+ve
for $i = 1, 2, 3, \dots, m$. As it is a cyclic network, $x_0 = x_m$.			

The inner product between the vector field characterized by the system dynamics within the hypercube and the normal to the outer facets of the hypercube is given in Table I. For protein concentrations between 0 and $\frac{\alpha}{\gamma}$, the value of Ξ is always positive. This implies that the vector field flows through the facets into the hypercube. Thus, the vector field flowing through all the outer facets is always directed into the hypercube, making the hypercube a positively invariant box.

Construction of an m - dimensional Toroidal-like Manifold: Trajectories approach the unique steady state along the path defined by the eigenvector of the negative eigenvalue. We calculated the eigenvector for the real negative eigenvalue. Let the eigenvector be $[\hat{x}_1 \ \hat{x}_2 \ \hat{x}_3 \ \dots \ \hat{x}_m]^T$. So,

$$\begin{bmatrix} -\gamma & 0 & 0 & \dots & -\rho \\ -\rho & -\gamma & 0 & \dots & 0 \\ 0 & -\rho & -\gamma & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -\gamma \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \vdots \\ \hat{x}_m \end{bmatrix} = -(\gamma + \rho) \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \vdots \\ \hat{x}_m \end{bmatrix},$$

$$\Rightarrow \hat{x}_m = \hat{x}_1, \hat{x}_1 = \hat{x}_2, \dots, \hat{x}_{m-1} = \hat{x}_m,$$

$$\Rightarrow [\hat{x}_1 \ \hat{x}_2 \ \dots \ \hat{x}_m]^T = [1 \ 1 \ \dots \ 1]^T. \quad (3)$$

In the direction of the eigenvector $[\hat{x}_1 \ \hat{x}_2 \ \hat{x}_3 \ \dots \ \hat{x}_m]^T$, we cut out a hypervolume from the given hypercube such that the steady state and the trajectories approaching it along this singular path are eliminated. This hypervolume may be visualized as a cylinder with a small radius $\varepsilon > 0$ along the eigenvector.

For some values of p , there is a possibility that some of these complex conjugate eigenvalues may have negative real part. So, we calculated the eigenvectors associated with such complex eigenvalues and removed the corresponding hypervolume along it. Let the eigenvector associated with such complex eigenvalues be $[\tilde{x}_1 \ \tilde{x}_2 \ \tilde{x}_3 \ \dots \ \tilde{x}_m]^T$. So,

$$\begin{bmatrix} -\gamma & 0 & 0 & \dots & -\rho \\ -\rho & -\gamma & 0 & \dots & 0 \\ 0 & -\rho & -\gamma & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -\gamma \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \vdots \\ \tilde{x}_m \end{bmatrix} = (-\gamma + \rho g(p, m)) \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \vdots \\ \tilde{x}_m \end{bmatrix},$$

where $g(p, m) = \left[\cos\left(\frac{2p+1}{m}\pi\right) + j \sin\left(\frac{2p+1}{m}\pi\right) \right]$ and

$$\begin{aligned} -\tilde{x}_m &= g(p, m) \tilde{x}_1, \\ -\tilde{x}_1 &= g(p, m) \tilde{x}_2, \\ &\vdots \\ -\tilde{x}_{m-1} &= g(p, m) \tilde{x}_m. \end{aligned} \quad (4)$$

Using De Moivre's theorem on (4), we get the eigenvector

$$\begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \vdots \\ \tilde{x}_m \end{bmatrix} = \begin{bmatrix} 1 \\ \cos\left(\frac{(2p+1)(m-1)}{m}\pi\right) + j \sin\left(\frac{(2p+1)(m-1)}{m}\pi\right) \\ -\cos\left(\frac{(2p+1)(m-2)}{m}\pi\right) - j \sin\left(\frac{(2p+1)(m-2)}{m}\pi\right) \\ \vdots \\ -\cos\left(\frac{2p+1}{m}\pi\right) - j \sin\left(\frac{2p+1}{m}\pi\right) \end{bmatrix}.$$

Although complex eigenvectors are not elements of the real space, their real and imaginary components reside within it. We divided the above eigenvector into its real and imaginary parts. These are the eigenvectors associated with a pair of complex conjugate eigenvalues,

$$v_{real} = \begin{bmatrix} 1 \\ \cos\left(\frac{(2p+1)(m-1)}{m}\pi\right) \\ -\cos\left(\frac{(2p+1)(m-2)}{m}\pi\right) \\ \vdots \\ -\cos\left(\frac{2p+1}{m}\pi\right) \end{bmatrix}, \quad v_{im} = \begin{bmatrix} 0 \\ \sin\left(\frac{(2p+1)(m-1)}{m}\pi\right) \\ -\sin\left(\frac{(2p+1)(m-2)}{m}\pi\right) \\ \vdots \\ -\sin\left(\frac{2p+1}{m}\pi\right) \end{bmatrix}.$$

Because some of these complex conjugate eigenvalues could have a negative real part, we removed the hypervolumes associated with the linear combination of v_{real} and v_{im} .

So, we obtained an m -dimensional toroidal-like manifold without the steady state. All the trajectories that enter this manifold stay inside because the hypercube is a positively invariant set and there are no trajectories that approach the steady state.

Continuous Map of a Cross-Section Onto Itself: We divided the hypercube into 2^m smaller hyperrectangles using the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$). Further, we computed the flow that traverses every facet within the smaller hyperrectangles.

Table II. Inner product between the vector field and the normal to the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$).

Hyperplane	$\hat{n}$	$\Xi = \langle \vec{G}, \hat{n} \rangle$	Value of Ξ
$x_i = x_0$	$\hat{x}_i$	$\frac{\alpha}{1 + \left(\frac{x_{i-1}}{k}\right)^n} - \gamma x_0$	$\begin{cases} +\text{ve, for } x_{i-1} < x_0, \\ -\text{ve, for } x_{i-1} > x_0. \end{cases}$
for $i = 1, 2, 3, \dots, m$. As it is a cyclic network, $x_0 = x_m$.			

The flow of the vector field through the outer facets of the hypercube obtained in Table I, combined with the flow through the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$) obtained in Table II, is used to determine the effective flow from one smaller hyperrectangle to its adjacent ones. We considered one of the smaller hyperrectangles, $0 \leq x_1 \leq x_0, 0 \leq x_2 \leq x_0, \dots, 0 \leq x_m \leq x_0$, as Box 1. From Table I, we have the flow of the vector field from the facets $x_i = 0$ (for $i = 1, 2, 3, \dots, m$) into Box 1. From Table II, we have the flow of the vector field from the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$) out of Box 1 and into the adjacent boxes. Similarly, we calculated the effective flow of the vector field for the rest of the boxes.

Upon inspection of the effective flow from one hyperrectangle to another, we observed that the vector field formed a closed path within certain hyperrectangles. This showed that a cross section of these smaller hyperrectangles mapped onto itself. By Brouwer's fixed point theorem, there exists a fixed point, which is a limit cycle for the system under consideration. The above construction also provides a rigorous bound on the region within the toroidal-like manifold where the limit cycle exists. $\square$

To summarize, we obtained the region in the hypercube where the limit cycle exists. This is done by constructing a toroidal-like manifold within a positively invariant set obtained by eliminating a hypervolume containing the steady state and the trajectories approaching it. Finally, we used Brouwer's Fixed Point theorem on a cross section of this manifold, which map onto itself, to show the existence of limit cycle.

B. Case Study

The differential equations describing the dynamics of a 5-gene cyclic regulatory network are given in (5).

$$\frac{dx_i}{dt} = F(x_{i-1}) - \gamma x_i, \text{ for } i = 1, 2, 3, \dots, 5. \quad (5)$$

Here x_i is the concentration of the i -th protein with $0 \leq x_i \leq \frac{\alpha}{\gamma}$. α, γ, k , and n are positive constants. $F(p) = \frac{\alpha}{1 + \left(\frac{p}{k}\right)^n}$ is the Hill function and n is the Hill coefficient. The eigenvalues of the Jacobian after the linearization of the system in (5) at the

steady state $(x_0, x_0, x_0, x_0, x_0)$ is $\lambda_{1,2,\dots,5} = -(\gamma + \rho), -\gamma + \rho(-0.31 \pm j0.95), -\gamma + \rho(0.81 \pm j0.59)$. The 5-gene regulatory network has a negative real eigenvalue. Along with it, one of the complex conjugate eigenvalues also has a negative real part. Guided by Theorem 1, we illustrated the existence of the limit cycle by considering the case where one pair of the complex conjugate eigenvalues has a positive real part.

Construction of an invariant box: We calculated the flow of the vector field through the different facets of the hypercube $0 \leq x_1, x_2, \dots, x_5 \leq \frac{\alpha}{\gamma}$. This is calculated along the unit normal vector of the various facets directed into the hypercube. Using the value of Ξ , we determined the flow of the vector field through the various facets and checked if the hypercube constituted a positively invariant set.

The vector field described by the system dynamics in (5) is: $\vec{G}(x_1, x_2, x_3, x_4, x_5) = [F(x_5) - \gamma x_1]\hat{x}_1 + [F(x_1) - \gamma x_2]\hat{x}_2 + [F(x_2) - \gamma x_3]\hat{x}_3 + [F(x_3) - \gamma x_4]\hat{x}_4 + [F(x_4) - \gamma x_5]\hat{x}_5$, where $\hat{x}_i$ (for $i = 1, 2, \dots, 5$) are the unit vectors in the direction of x_i axis. The value of the inner product between the vector field and the normal to the facets directed into the hypercube is obtained from Table I by substituting $m = 5$. The values of Ξ are positive for every outer facet of the hypercube. This indicated that the flow of the vector field described by the system dynamics is always directed inward and stays inside the hypercube, making it a positively invariant set.

Construction of a 5-dimensional Toroidal-like Manifold: The existence of a unique steady state in the given hypercube indicated that certain trajectories approach the steady state along the singular path defined by the eigenvector of the negative eigenvalue. So, we calculated the eigenvector for the real negative eigenvalue. Let the eigenvector be $[\hat{x}_1 \ \hat{x}_2 \ \hat{x}_3 \ \hat{x}_4 \ \hat{x}_5]^T$. So, using (3) the eigenvector is found to be $[1 \ 1 \ 1 \ 1 \ 1]^T$. In the direction of this eigenvector we cut out a hypervolume from the given hypercube, such that the steady state and the trajectories approaching it along this singular path are eliminated.

The regulatory network also has a pair of complex conjugate eigenvalues with negative real part. So, we calculated the eigenvectors associated with these complex eigenvalues and removed a hypervolume along it. Using the relation in (4) and the De Moivre's theorem, we get the eigenvector as: $[\tilde{x}_1 \ \tilde{x}_2 \ \tilde{x}_3 \ \tilde{x}_4 \ \tilde{x}_5]^T = [1 \ 0.31 + j0.95 \ -0.81 + j0.59 \ -0.81 - j0.59 \ 0.31 - j0.95]^T$. We divide the above eigenvector into its real and imaginary parts. These are the eigenvectors associated complex eigenvalue with negative real part: $v_{real} = [1 \ 0.31 \ -0.81 \ -0.81 \ 0.31]^T$ and $v_{im} = [0 \ 0.95 \ 0.59 \ -0.59 \ -0.95]^T$. The complex conjugate eigenvalues produce conjugate eigenvectors. So, the v_{real} remains same for the complex conjugate pair. v_{im} for the pair of eigenvalues is a conjugate pair. However, the conjugate gives the same eigenvector but in the opposite direction. In the sense of removing a hypervolume, the v_{im} can be considered to be same. So, we also removed the hypervolume along the eigenvectors v_{real} and v_{im} .

So, we obtained a toroidal-like manifold with no steady state. All the trajectories are inside the torus-like structure

(because the hypercube is a positively invariant set).

Table III. Direction of flow of vector field within the smaller boxes.

Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
12	$0 \leq x_1 \leq x_0, x_0 \leq x_2 \leq \frac{\alpha}{\gamma}, 0 \leq x_3 \leq x_0,$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}, x_0 \leq x_5 \leq \frac{\alpha}{\gamma}.$	Box 12 $\rightarrow$ Box 24
13	$0 \leq x_1 \leq x_0, x_0 \leq x_2 \leq \frac{\alpha}{\gamma}, x_0 \leq x_3 \leq \frac{\alpha}{\gamma},$ $0 \leq x_4 \leq x_0, x_0 \leq x_5 \leq \frac{\alpha}{\gamma}.$	Box 13 $\rightarrow$ Box 23
16	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}, 0 \leq x_2 \leq x_0, x_0 \leq x_3 \leq \frac{\alpha}{\gamma},$ $0 \leq x_4 \leq x_0, x_0 \leq x_5 \leq \frac{\alpha}{\gamma}.$	Box 16 $\rightarrow$ Box 21
17	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}, 0 \leq x_2 \leq x_0, x_0 \leq x_3 \leq \frac{\alpha}{\gamma},$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}, 0 \leq x_5 \leq x_0.$	Box 17 $\rightarrow$ Box 28
19	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}, x_0 \leq x_2 \leq \frac{\alpha}{\gamma}, 0 \leq x_3 \leq x_0,$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}, 0 \leq x_5 \leq x_0.$	Box 19 $\rightarrow$ Box 26
21	$0 \leq x_1 \leq x_0, 0 \leq x_2 \leq x_0, x_0 \leq x_3 \leq \frac{\alpha}{\gamma},$ $0 \leq x_4 \leq x_0, x_0 \leq x_5 \leq \frac{\alpha}{\gamma}.$	Box 21 $\rightarrow$ Box 13
23	$0 \leq x_1 \leq x_0, x_0 \leq x_2 \leq \frac{\alpha}{\gamma}, 0 \leq x_3 \leq x_0,$ $0 \leq x_4 \leq x_0, x_0 \leq x_5 \leq \frac{\alpha}{\gamma}.$	Box 23 $\rightarrow$ Box 12
24	$0 \leq x_1 \leq x_0, x_0 \leq x_2 \leq \frac{\alpha}{\gamma}, 0 \leq x_3 \leq x_0,$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}, 0 \leq x_5 \leq x_0.$	Box 24 $\rightarrow$ Box 19
26	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}, 0 \leq x_2 \leq x_0, 0 \leq x_3 \leq x_0,$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}, 0 \leq x_5 \leq x_0.$	Box 26 $\rightarrow$ Box 17
28	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}, 0 \leq x_2 \leq x_0, x_0 \leq x_3 \leq \frac{\alpha}{\gamma},$ $0 \leq x_4 \leq x_0, 0 \leq x_5 \leq x_0.$	Box 28 $\rightarrow$ Box 16
Complete table in Supplementary Material Part I.		

Continuous Map of a Cross-Section Onto Itself: The hypercube is divided into 2^5 smaller hyperrectangles by using the hyperplanes $x_1 = x_0, x_2 = x_0, \dots$, and $x_5 = x_0$. We obtained the value of Ξ between the vector field describing the system dynamics and the unit normal vector through these hyperplanes from Table II by substituting $m = 5$.

Using the values of Ξ between (a) the vector field and the normal to the outer facets of the hypercube and (b) the vector field and the normal to the hyperplanes $x_i = x_0$ (for $i = 1, 2, \dots, 5$), we determined the direction of flow of the vector field within the smaller boxes. This is summarized in Table III (For complete table refer to Supplementary Material). Upon inspection of Table III, we find that the vector field formed a continuous map of a cross section onto itself. The continuous map is as follows: $12 \rightarrow 24 \rightarrow 19 \rightarrow 26 \rightarrow 17 \rightarrow 28 \rightarrow 16 \rightarrow 21 \rightarrow 13 \rightarrow 23 \rightarrow 12$. We obtained a toroidal-like manifold (which is a positively invariant set) and a cross section of this structure formed a continuous map onto itself. So, by Brouwer's Fixed Point theorem, there exists a limit cycle.

III. LOCALIZATION

The Brouwer's Fixed Point theorem guarantees the presence of a fixed point and thus a periodic orbit through a cross-section that maps onto itself. However, where the limit cycle trajectory lies is still undetermined. Thus localization is the natural next step in rigorously refining the region of existence of the limit cycle. We now use an interval-based Reachability

Analysis²⁵ to localize the region of existence. This method is based on rigorously solving the network dynamics using Interval Analysis. We consider a set of initial conditions within the region of existence that we want to localize. By the definition of limit cycle, the sets obtained using Reachability Analysis that do not map onto themselves, under the flow of the system dynamics, does not contain a limit cycle. Therefore the limit cycle can be localized by combining the existence result from the Brouwer's Fixed Point theorem with the non-existence criterion implemented by the set-based Reachability Analysis.

A. Methodology

The methodology involves dividing the region of existence into smaller hyperrectangles, evolving the dynamics from these smaller hyperrectangles through Reachability Analysis (flowpipes), and examining the 'return map'. Each region is categorized as: (i) Does not contain the limit cycle, (ii) May contain the limit cycle, or (iii) Contains the limit cycle. This classification fundamentally depends on a geometric subset test to ascertain whether the rigorously computed return set X_r is entirely encompassed within the original set X_0 .

This interval-based Reachability Analysis approach thus serves as a complement of the Brouwer's Fixed Point theorem's topological arguments using validated numerical techniques. In this approach, the geometric confinement test extends the theoretical proof by connecting the existence result to a rigorous characterization of the system's oscillatory dynamics.

B. Case Study

The existence proof for the three-gene cyclic regulatory network within an invariant cube $0 \leq x_1, x_2, x_3 \leq \frac{\alpha}{\gamma}$, when divided into 2^3 smaller boxes by the surfaces $x_i = x_0$ (for $i = 1, 2, 3$), gives us six cross-sections of the smaller boxes that map onto themselves. We then illustrated the interval-based Reachability Analysis to localize periodic orbit through one of these cross-sections that map onto themselves in the three-gene cyclic regulatory network. The framework successfully divided the cross-section according to the underlying dynamical behaviour of the three-gene cyclic regulatory network. The methodology, based on partitioning the cross-section and analyzing the flowpipe of each partition, showed several findings, which are visually summarized in Fig. 1 (For complete details about methodology refer to Supplementary Material Part II).

The analysis classified the vast majority of the partitioned region of this cross-section as not containing a periodic orbit. The black box in Figure 1, contains the unstable steady state (x_0, x_0, x_0) (the value of x_0 for this simulation is 5.361) of the system and is excluded from the Reachability Analysis. A distinct cluster of initial sets, marked in yellow in Figure 1, was classified as "May contain the limit cycle". A cluster of green boxes around the box containing the steady state is classified as "Failed boxes". This is because of the failure of the reachability analysis to return a solution. This can be attributed to

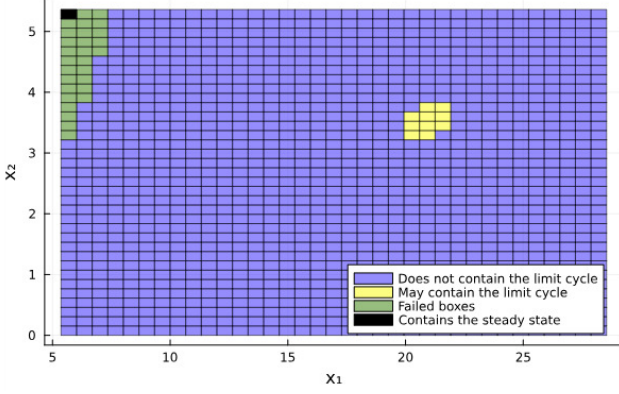


Figure 1. Projection of the cross-section, that maps onto itself, showing the geometric classification results obtained from the Reachability Analysis. The system parameters used in the simulation are: $\alpha = 26.26$, $\gamma = 0.92$, $k = 4$, $n = 5$.

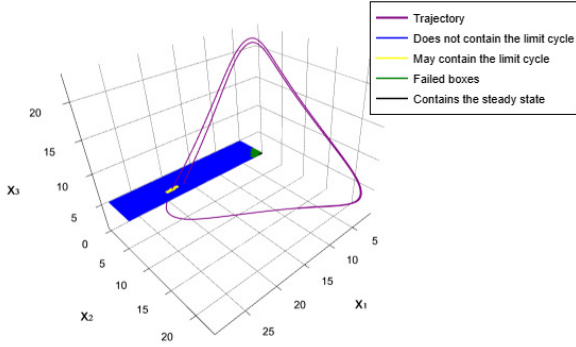


Figure 2. The numerical solution of the three-gene cyclic regulatory network with initial condition $(19, 3.5, 5.361)$ passes through the region classified as “May contain the limit cycle” in Fig. 1. The system parameters used in the simulation are: $\alpha = 26.26$, $\gamma = 0.92$, $k = 4$, $n = 5$.

solver crash or blow up in the bounds due to overapproximation. Further in Fig. 2 the numerical solution of the three gene cyclic regulatory network model is plotted along with one of the cross-section that maps onto itself. Fig. 3 provides a magnified view of the trajectory passing through the region classified by the method as “May contain the limit cycle”. This finding validates the method’s efficacy in localizing the region in which the limit cycle exists.

IV. STABILITY ANALYSIS

Having established the existence of a limit cycle within an invariant hypercube (after removing the unstable steady state along its stable manifolds) using Brouwer’s Fixed Point theorem, we then localize the limit cycle along the cross-section that maps onto itself using an interval based Reachability Analysis. We proceed to prove the stability of the limit cycle that exists within this localized region.

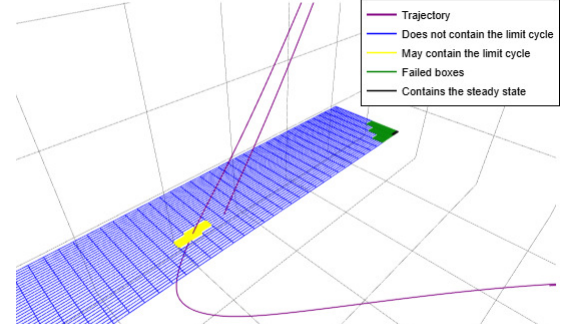


Figure 3. A magnified view of Fig. 2.

A. Methodology

The Poincaré Map as a Stability Proof: To evaluate the stability of the limit cycle, we discretize the continuous-time dynamics using a Poincaré section Σ , defined as a $(m - 1)$ -dimensional transversal surface (e.g., the cross-section which maps onto itself). The classical Poincaré map $P : \Sigma \rightarrow \Sigma$ is a discrete-time function that maps an initial condition on this surface to its subsequent intersection after one full period T ²⁶. The stability proof requires locating the exact analytical fixed point $x^* = P(x^*)$ on this section¹. If the eigenvalues of the Jacobian matrix of the discrete Poincaré Map evaluated at x^* lie strictly inside the unit circle, the limit cycle is stable²⁷. However, finding the exact analytical coordinates, i.e., the standard point based analysis for the highly nonlinear biological oscillators is difficult.

To address this, we generalize the Poincaré map from discrete points to intervals using Reachability Analysis. Instead of tracking a single trajectory, we define a $(m - 1)$ -dimensional interval region $X \subset \Sigma$ and evolve it as a continuous bounding flowpipe. By studying the behavior of the trajectories over the entire flowpipe, we verify that the Poincaré map contracts. This would confirm the stability of the limit cycle within the previously localized region, without requiring its exact coordinates.

During integration, this continuous flowpipe is represented as a discrete sequence of sets (Taylor models or zonotopes)²⁵. When the flowpipe completes a period and intersects the section Σ , multiple consecutive sets cross the plane over a window of time. To guarantee the rigorous enclosure of the discrete return map $P(X)$, we aggregate this entire sequence of intersecting sets and compute their intersection. This produces a single bounding box that strictly contains the set that returns to Σ .

The Interval Return Map Derivative: We now establish the notations for the exact and interval dynamics:

- Let Σ denote the Poincaré section which is an $(m - 1)$ -dimensional transversal surface, i.e., the cross-section that maps onto itself.
- Let $x \in \mathbb{R}^m$ be an initial state vector defined on the vector field in (1), and $X \subset \Sigma$ be a bounded interval set of initial conditions such that $x \in X$.

- Let $h(x)$ be a scalar constraint function that defines the Poincaré section, such that for any point on the section $h(x) = 0$.
- Let $\nabla h(x)$ be the gradient of this constraint function and is the normal vector n pointing perpendicular to Σ .
- Let $\phi(t, x)$ represent the continuous-time flow. Its set-based counterpart, $\phi(t, X)$, is the continuous flowpipe bounding all the trajectories originating from X .
- Let $T(x)$ denote the exact analytical return time function required for a point trajectory to intersect the section Σ again. For the bounded set X , the return time is strictly bounded by the interval window $T(X) = [t_{\min}, t_{\max}]$, representing the times the first and last sets intersect Σ respectively.
- Let $P(x)$ represent the discrete Poincaré map for a point, constructed by evaluating the flow at the return time: $P(x) = \phi(T(x), x)$. Its interval extension, $P(X)$, is bounded by evaluating the flowpipe over the entire return time window $[t_{\min}, t_{\max}]$.

To computationally evaluate the contraction of this discrete interval-based Poincaré map, we must rigorously derive an expression for its Jacobian matrix, denoted as $DP(x)$, and then compute its interval extension over the set X .

Derivative using the Chain Rule: To find the Jacobian matrix of the Poincaré map, $DP(x) = \frac{d}{dx}P(x)$, we must differentiate the function $P(x) = \phi(T(x), x)$ with respect to the initial state x . Because the initial state x dictates both the spatial trajectory and the total time taken to return to the section, ϕ depends on x through both of its arguments. Applying the chain rule, the total derivative expands into two distinct partial derivatives:

$$\frac{d}{dx}\phi(T(x), x) = \left. \frac{\partial \phi(t, x)}{\partial t} \right|_{t=T(x)} \frac{\partial T(x)}{\partial x} + \left. \frac{\partial \phi(t, x)}{\partial x} \right|_{t=T(x)} \frac{\partial x}{\partial x}$$

We evaluate each component of this expansion individually:

1. The partial derivative $\frac{\partial \phi}{\partial t}$ is simply the definition of our continuous vector field, $f(\phi(t, x))$. Evaluated at the return time $t = T(x)$, this becomes the velocity vector at the point of intersection: $f(P(x))$.
2. The time gradient $\frac{\partial T(x)}{\partial x}$ represents how the scalar return time changes with respect to the m -dimensional initial state. The derivative of a scalar function with respect to a column vector yields a $1 \times m$ Jacobian row vector. We denote this as the transpose of the gradient, $\nabla T(x)^T = \left[\frac{\partial T}{\partial x_1}, \dots, \frac{\partial T}{\partial x_m} \right]$.
3. The partial derivative $\frac{\partial \phi}{\partial x}$ is the fundamental solution (or monodromy) matrix, $\Phi(t, x)$, which tracks how small perturbations around a trajectory grow, shrink, or change direction over time. Evaluated at $t = T(x)$, this becomes $\Phi(T(x), x)$.
4. The derivative of the initial state with respect to itself, $\frac{\partial x}{\partial x}$, trivially reduces to the identity matrix, I_m .

Substituting these evaluated components back into our total derivative equation yields the explicit chain rule formulation:

$$DP(x) = f(P(x))\nabla T(x)^T + \Phi(T(x), x) \quad (6)$$

Enforcing the Transversality Condition: In (6), the return time gradient $\nabla T(x)^T$ is unknown and must be eliminated. This is done by using the geometric invariance of the Poincaré map, i.e., the returned point $P(x)$ must always strictly lie exactly on the section Σ . Therefore, the constraint function evaluated at the return point is $h(P(x)) = 0$. Because this relationship holds across the entire section, differentiating this scalar equation with respect to the m -dimensional vector x must yield a $1 \times m$ row vector of zeros ($\mathbf{0}^T$). Applying the chain rule to $h(P(x))$, we differentiate the outer scalar function h with respect to its spatial input y and multiply it by the Jacobian of the inner mapping P :

$$\left. \frac{\partial h(y)}{\partial y} \right|_{y=P(x)} \frac{dP(x)}{dx} = \mathbf{0}^T$$

The term $\frac{\partial h}{\partial y}$ in the chain rule expression represents the total derivative of a scalar function with respect to a column vector. This forms a $1 \times m$ Jacobian row vector containing the partial derivatives. Geometrically, the vector strictly perpendicular to $h(y) = 0$ is the normal vector n . This normal vector is defined by the gradient, $n = \nabla h(y)$, which is expressed as an $m \times 1$ column vector. Because both vectors are composed of the exact same partial derivatives, the row vector is simply the transpose of the geometric normal column vector: $\frac{\partial h}{\partial y} = n^T = \nabla h(y)^T$.

Evaluating this transposed normal vector at the return point $y = P(x)$ gives $\nabla h(P(x))^T$. The inner derivative $\frac{dP(x)}{dx}$ is the $m \times m$ Jacobian matrix $DP(x)$. So, we substitute these terms back into the chain rule validating the matrix dimensions and produce the final transversality condition:

$$\nabla h(P(x))^T DP(x) = \mathbf{0}^T \quad (7)$$

Isolating the Return Time Gradient: We substitute the derived expression for $DP(x)$ from (6) into the transversality condition in (7):

$$\begin{aligned} & \nabla h(P(x))^T \left[f(P(x))\nabla T(x)^T + \Phi(T(x), x) \right] = \mathbf{0}^T \\ \implies & \nabla h(P(x))^T f(P(x))\nabla T(x)^T + \nabla h(P(x))^T \Phi(T(x), x) = \mathbf{0}^T \\ \implies & \left(\nabla h(P(x))^T f(P(x)) \right) \nabla T(x)^T = -\nabla h(P(x))^T \Phi(T(x), x) \\ \implies & \nabla T(x)^T = -\frac{\nabla h(P(x))^T \Phi(T(x), x)}{\nabla h(P(x))^T f(P(x))} \end{aligned} \quad (8)$$

Now, for $\nabla T(x)^T$ to exist, the denominator is non-zero. So,

$$\nabla h(P(x))^T f(P(x)) \neq 0. \quad (9)$$

Geometrically the denominator being zero signifies the vector field $F(P(x))$ being tangent to the Poincaré section at the return point.

Assembling the Projection Matrix: With the time gradient explicitly defined in (8), we substitute it back into the original chain rule equation (6):

$$DP(x) = f(P(x)) \left[-\frac{\nabla h(P(x))^T \Phi(T(x), x)}{\nabla h(P(x))^T f(P(x))} \right] + \Phi(T(x), x)$$

In the first term, multiplying the $m \times 1$ column vector $f(P(x))$ by the $1 \times m$ row vector $\nabla h(P(x))^T$ executes an outer product, producing an $m \times m$ matrix. Factoring out $\Phi(T(x), x)$ to the right-hand side, we obtain the finalized closed-form projection matrix for the Poincaré derivative:

$$DP(x) = \left[I_m - \frac{f(P(x)) \nabla h(P(x))^T}{\nabla h(P(x))^T f(P(x))} \right] \Phi(T(x), x) \quad (10)$$

The bracketed term in (10) acts as a geometric projection operator.

Evaluating the Derivative over Interval Sets: The projection formula in (10) is mathematically exact for a point x . To calculate the derivative across the entire bounded set X , we utilize the concept of inclusion function²⁸. Interval Arithmetic guarantees that replacing exact real variables in an analytical formula with their corresponding interval bounds produces a comprehensive inclusion function that strictly contains all possible point-wise evaluations.

We substitute the exact variables with their corresponding intervals evaluated over the crossing window $T(X) = [t_{\min}, t_{\max}]$:

- The exact flow vector $f(P(x))$ is replaced by the interval vector $\mathbf{f}^I$, which strictly contains all possible velocities of the vector field within all the sets intersecting Σ between interval window $T(X)$.
- The exact continuous variation matrix $\Phi(T(x), x)$ is replaced by the interval matrix Φ^I , which encompasses the continuous monodromy matrices for all time steps $t \in [t_{\min}, t_{\max}]$.
- The normal vector $n = \nabla h(x)$ remains constant.

By substituting these intervals directly into the projection formula in (10), we compute an interval matrix $\mathbf{DP}^I$:

$$\mathbf{DP}^I = \left[I_m - \frac{\mathbf{f}^I n^T}{n^T \mathbf{f}^I} \right] \Phi^I$$

This interval matrix rigorously bounds the Jacobian $DP(x)$ for every possible trajectory originating within the initial set X , transitioning point-wise calculations into a set-based calculations.

Computation of the Monodromy Matrix and The Augmented System: To obtain the continuous variation matrix $\Phi(T(x), x)$ dynamically, it must be integrated alongside the system's state space. $\Phi(t, x)$ evolves according to the matrix differential equation:

$$\dot{\Phi} = J(\phi(t, x))\Phi, \quad \Phi(0) = I_{m \times m}$$

where $J(x)$ is the state-dependent Jacobian of the vector field $\dot{x} = F(x)$. For the cyclic gene regulatory network in (2), the

Jacobian matrix is defined as:

$$J(x) = \begin{bmatrix} -\gamma & 0 & \dots & 0 & -\rho(x_m) \\ -\rho(x_1) & -\gamma & \dots & 0 & 0 \\ 0 & -\rho(x_2) & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & -\gamma & 0 \\ 0 & 0 & \dots & -\rho(x_{m-1}) & -\gamma \end{bmatrix}$$

To track the spatial variation across the entire interval set, we formulate a $(m + m^2)$ -dimensional augmented system. This system couples the m primary state variables representing the continuous vector field in (2) with the m^2 elements of the $m \times m$ monodromy matrix (Φ_{ij}) , which evolve according to $\dot{\Phi} = J(x)\Phi$. Expanding this matrix multiplication yields the explicit differential equations for each individual element of the monodromy matrix:

For any given column $j \in \{1, 2, \dots, m\}$ of the monodromy matrix, the elements evolve according to:

$$\begin{aligned} \dot{\Phi}_{1j} &= -\gamma \Phi_{1j} + -\rho(x_m) \Phi_{mj} \\ \dot{\Phi}_{ij} &= -\rho(x_{i-1}) \Phi_{i-1,j} - \gamma \Phi_{ij} \quad \text{for } i = 2, 3, \dots, m \end{aligned}$$

By integrating this $(m + m^2)$ -dimensional augmented system from the initial interval state $(X, I_{m \times m})$ to the validated return window $T(X)$, we extract the exact interval bounds for the continuous variation matrix Φ^I .

B. Algorithm and Computational Implementation

The theoretical derivation for the stability proof using Poincaré map is computationally implemented in Julia using interval-based Reachability Analysis. Based on the above framework, the computational proof is executed via Algorithm 1.

As established in the previous work, stability is confirmed if the exact eigenvalues of the Jacobian $DP(x)$ lie within the unit circle ($|\lambda_{\max}| < 1$)²⁷. Computing the exact eigenvalues for interval matrices is prone to overestimation. So, we utilize the Spectral Radius Theorem, which guarantees that the maximum eigenvalue is strictly bounded by any induced matrix norm ($|\lambda_{\max}| \leq \|A\|$)²⁹. Therefore, Algorithm 1 calculates the interval infinity-norm q across the entire discrete returning reachable sets by taking the maximum row sum of the interval Jacobian $DP(X)$:

$$q = \sup_{x \in X} \|DP(x)\|_{\infty}$$

If $q < 1$, the worst-case stretching along the grid axes is strictly bounded. This mathematically guarantees that the theoretical eigenvalues of the interval map are trapped safely inside the unit circle, confirming the stability of the limit cycle. The condition $q < 1$ rigorously confirms that the phase space volume is strictly contracting transversally. Supported by the preceding existence proof and localization, this interval-based Poincaré Map computation proves the limit cycle as stable.

Algorithm 1 Interval-Based Poincaré Map Stability Proof

Require: Set of candidate hyperrectangles on Σ defined by $h(x) = 0$, normal vector $n = \nabla h(x)$, system parameters, $T_{\max}$

Ensure: List of strictly stable sub-elements $\mathcal{S}$

```

1:  $\mathcal{S} \leftarrow \emptyset$ 
2: Subdivide initial candidate bounding boxes into a finer interval
   grid  $\mathcal{G}$  on  $\Sigma$ 
3: for each sub-element  $X \in \mathcal{G}$  do
4:   Phase 1: Reachability Analysis
5:   Integrate continuous  $m$ -dimensional flowpipe  $\phi(t, X)$  gov-
   erned by  $\dot{x} = f(x)$ 
6:   if  $\phi(t, X)$  does not geometrically intersect roughly the origi-
   nal bounds on  $\Sigma$  then
7:     Discard sub-element  $X$ 
8:   else
9:     Phase 2: Poincaré Map Analysis
10:    Initialize  $(m + m^2)$ -dimensional augmented state:  $Z_0 \leftarrow$ 
       $(X, I_{m \times m})$ 
11:    Integrate  $(m + m^2)$ -dimensional system  $\dot{\Phi} = J(\phi(t, X))\Phi$ 
      up to  $T_{\max}$  tracking interval state and  $\Phi$ 
12:    Extract validated interval return time window  $T(X)$  and
      return intersection bounding box
13:    if transversality condition  $(n^T \mathbf{f}^t \neq 0)$  fails across the
      crossing bundle then
14:      Discard sub-element  $X$ 
15:    else
16:      Extract  $m \times m$  interval variational matrix  $\Phi^I$  and flow
      vector  $\mathbf{f}^t$  at intersection
17:      Compute  $m$ -dimensional closed-form projection:
       $\mathbf{DP}_{mD}^I \leftarrow \left[ I_m - \frac{\mathbf{f}^t n^T}{n^T \mathbf{f}^t} \right] \Phi^I$ 
18:      Restrict to  $(m - 1)$ -dimensional tangent space of  $\Sigma$ :
       $\mathbf{DP}^I \leftarrow \mathbf{DP}_{mD}^I[1 : m - 1, 1 : m - 1]$ 
19:      Compute interval infinity-norm bound:  $q \leftarrow$ 
       $\sup_{x \in X} \|\mathbf{DP}^I\|_{\infty}$ 
20:      if  $q < 1.0$  then
21:         $\mathcal{S} \leftarrow \mathcal{S} \cup \{X\}$  ▷ Contraction verified
22:      end if
23:    end if
24:  end if
25: end for
26: return  $\mathcal{S}$ 

```

C. Case Study

To execute the stability proof, the computational framework given in Algorithm 1 was applied to 3-gene cyclic regulatory network with the following parameters: $\alpha = 26.26$, $\gamma = 0.92$, and $n = 5$. The Poincaré section was defined at the cross-section that maps onto itself, constrained by the scalar function $h(x) = x_3 - x_0 = 0$

Following the localization of the cross-section, the classified grid elements were subdivided and evaluated using Algorithm 1. The resulting projection of the Poincaré section is illustrated in Figure 4. The yellow regions in Figure 4 denote the region in the cross-section where the limit cycle may lie. Upon finer subdivision, the algorithm successfully isolated a cluster of sub-elements where the interval infinity-norm of the projected discrete Jacobian $DP(X)$ was bounded strictly below 1.0, satisfying the stability condition, high-

lighted in cyan. The rigorous upper bounds on the induced norm, $q = \sup \|DP(X)\|_{\infty}$, for these certified sub-elements are detailed in Table IV. Because $q < 1.0$ across this localized grid, the Poincaré Map contracts and the limit cycle is stable within this region. To validate this stability check, a traditional ODE solver was used to simulate the numerical solution of the differential model. The intersection of this periodic orbit with the Poincaré section is plotted as the black marker. This intersection lies within the verified cyan sub-elements.

Table IV. Certified Stable Sub-elements on Σ ($x_3 = x_0$) and their q values (Cyan Boxes in Figure 4)

Sub-element Index (Cyan Boxes)	Interval Bounds on Σ ($X_1 \times X_2$)	Contraction Bound (q)
Box 1	$[20.94, 21.03] \times [3.47, 3.50]$	0.38
Box 2	$[20.75, 20.85] \times [3.50, 3.52]$	0.39
Box 3	$[20.85, 20.94] \times [3.50, 3.52]$	0.39
Box 4	$[20.94, 21.03] \times [3.50, 3.52]$	0.39
Box 5	$[21.03, 21.13] \times [3.50, 3.52]$	0.38
Box 6	$[20.75, 20.85] \times [3.52, 3.54]$	0.39
Box 7	$[20.85, 20.94] \times [3.52, 3.54]$	0.39
Box 8	$[20.94, 21.03] \times [3.52, 3.54]$	0.39
Box 9	$[21.03, 21.13] \times [3.52, 3.54]$	0.38
Box 10	$[20.85, 20.94] \times [3.54, 3.56]$	0.40
Box 11	$[20.94, 21.03] \times [3.54, 3.56]$	0.39
Box 12	$[21.03, 21.13] \times [3.54, 3.56]$	0.38

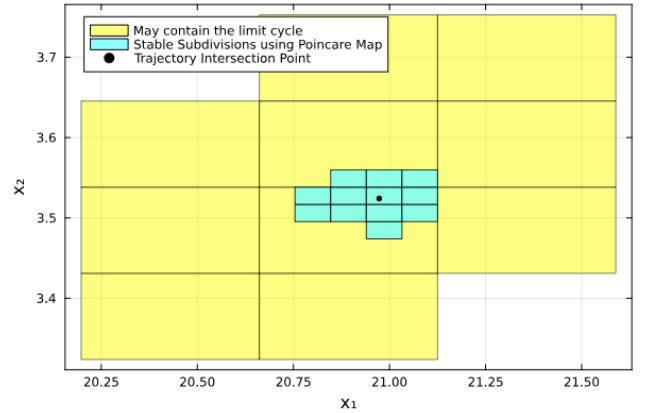


Figure 4. Further classification of the localized cross-section using Poincaré Map to localize the region where the stable limit cycle lies. Yellow regions represent the grid elements that may contain the limit cycle from the previous localization using Reachability Analysis. Cyan regions highlight the rigorously verified sub-elements where $\|DP(x)\|_{\infty} < 1$ is guaranteed. The black marker denotes the crossing of the numerical solution at the cross-section.

V. CONCLUSION

We proved the existence of a limit cycle in a class of cyclic gene regulatory networks and provided a rigorous localiza-

tion of this region of existence. The existence proof, based on Brouwer's Fixed Point theorem, is an alternative and elementary proof for this class of systems, with the benefit of an enhanced geometric visualisation. This rigorous localization relies on an interval-based Reachability Analysis, which, in conjunction with Brouwer's Fixed Point theorem, can provide a significantly tighter enclosure of the limit cycle trajectory. The localized region is analyzed using an interval-based Poincaré Map to verify contraction and prove the limit cycle is stable.

For the existence proof, we constructed a positively invariant set for the network dynamics by exploiting the inherent boundedness and by creating a toroidal-like manifold region that eliminates hypervolume consisting of the unstable steady state and its associated stable manifolds. We found that a cross section of this manifold continuously maps onto itself under the system dynamics, showing via Brouwer's Fixed Point theorem, the existence of a limit cycle. These results are illustrated using a 5-dimensional example. After establishing the existence of a limit cycle in cyclic gene regulatory networks, we employed interval-based Reachability Analysis to localize the region of its existence across the cross-sections that map onto themselves. The Reachability Analysis was demonstrated using a 3-dimensional example, offering a visual representation of the categorized regions. The stability of the localized limit cycle was established using an interval-based Poincaré Map analysis. The Jacobian of the discrete Poincaré Map was calculated over the localized cross-section using the variational dynamics. The sub-elements, where the eigenvalues of the Jacobian of the map lay within the unit circle, were identified. This guaranteed that the corresponding Poincaré Map is locally contractive and, subsequently establishes the stability of the limit cycle within the verified region. The stability proof using the Poincaré Map was verified for the previously localized region of the 3-dimensional example.

This work establishes a workflow for analyzing oscillatory dynamics in nonlinear biomolecular systems. The methodology provides an elegant framework for demonstrating the existence of limit cycles, rigorously localizing the region of existence, and verifying their stability through an interval-based Poincaré Map analysis.

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SUPPLEMENTARY MATERIAL

The supplementary material includes the complete Table III and the methodology of the Reachability Analysis for the three-gene cyclic regulatory network.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Sidhanta Mohanty: Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. Shaunak Sen: Conceptualization, Formal analysis, Investigation, Supervision, Writing – original draft, Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT (OpenAI) to partially generate the LaTeX syntax and formatting for the algorithms and to rectify syntax errors and debug error messages in some sections of the code available in the accompanying GitHub Repository. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the contents of the work.

DATA AVAILABILITY

The algorithms used for generating the plots in this study are available within the supplementary material. The code used to localize and verify stability of the limit cycle via interval-based Reachability Analysis and Poincaré Map is provided at [GitHub Repository](#).

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